

Summary: *Do Tax Cuts Generate Twin Deficits? A Multi-Country Analysis*

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1 Paper Overview

1.1 Introduction & Context

The “twin deficits” debate asks whether government budget deficits cause external (current account) deficits. Large fiscal deficits—especially those arising from tax cuts—are hypothesised to reduce national saving and thereby worsen a country’s external position. However, the raw data for industrialised countries show no uniform positive comovement between budget and external deficits: some countries exhibit positive correlations, others negative, and many are statistically insignificant. This heterogeneity arises because the raw series jointly capture the effects of many different shocks (productivity, government spending, tax, etc.), which may push the two deficits in opposite directions simultaneously.

Boileau and Normandin address this identification problem by developing a structural approach based on a small open economy model with overlapping generations (OLG). Their framework isolates the effects of *tax shocks* specifically, thereby testing the twin deficit hypothesis against the Ricardian equivalence hypothesis. The paper’s distinguishing contribution is that the structural model nests two distinct channels—a *demographic channel* and a *forecasting channel*—through which the budget deficit can affect the external deficit. By empirically disentangling these channels across 16 industrialised countries over the post-1970 period, the authors provide a comprehensive, multi-country assessment of the twin deficit phenomenon.

1.2 Structure of the Paper

1. **Section 1 (Introduction):** Motivates the twin deficit question, surveys related literature, and previews the structural approach and main findings.
2. **Section 2 (A Small Open Economy):** Presents the Blanchard (1985) OLG model, derives the analytical solution (current account rule, adjusted current account, aggregate wealth dynamics), and discusses the demographic and forecasting channels.
3. **Section 3 (The Empirical Approach):** Generalises the model’s solution into a vector error correction model (VECM) framework, discusses estimation of the birth rate and model restrictions, and describes the Cholesky-based identification of tax shocks.
4. **Section 4 (Empirical Results):** Reports unconditional and conditional correlations, impulse responses, variance decompositions, and cumulative debt responses; disentangles the demographic and forecasting channels.
5. **Section 5 (Discussion):** Summarises findings, acknowledges model limitations (lump-sum taxes, exogenous interest rate, small-open-economy assumption), and suggests extensions.

1.3 Conclusions

The paper reaches several key conclusions. First, once the effects of other shocks are removed and only tax shocks are retained, the conditional correlations between budget and external deficits are uniformly positive, large, and statistically significant for all 16 countries—strong evidence for the twin deficit hypothesis. Second, both the demographic and forecasting channels play important roles: the demographic channel (wealth effect from intergenerational tax-burden shifting) explains the *amplitude* of the external deficit’s response to a tax cut, while the forecasting channel (budget deficit contains information useful for predicting future net output) explains the *shape* of the dynamic response. Third, the tax shock accounts for a substantial portion of the forecast error variance of both deficits, and generates economically meaningful cumulative increases in public and external debt over a decade.

1.4 Key Takeaways

1. **Tax cuts generate twin deficits across all 16 countries studied.** Conditioning on tax shocks, the correlations between budget and external deficits are large, positive, and significant—a sharp contrast with the heterogeneous unconditional correlations.
2. **Isolating tax shocks is essential.** Raw data show no clear pattern because multiple shocks (productivity, government spending, taxes) push the deficits in different directions simultaneously.
3. **Two channels transmit the budget deficit to the external deficit.** The *demographic channel* works through the OLG structure (consumers shift taxes to unborn generations), and the *forecasting channel* works through the informational content of the budget deficit for predicting future net output.
4. **The channels have distinct roles:** the forecasting channel shapes the dynamic path of the external deficit response, while the demographic channel determines its magnitude.
5. **The structural OLG model fits the data well** for all 16 countries, with actual birth rates falling within the bounds implied by the model's orthogonality restrictions.

2 Key Points

2.1 The Twin Deficit Hypothesis versus Ricardian Equivalence

The paper contrasts two classical hypotheses about the effect of a tax cut on the external deficit. The *twin deficit hypothesis* stipulates that a tax cut raises both the budget deficit and the external deficit, because reduced taxes lower national saving and increase foreign borrowing. In contrast, the *Ricardian equivalence hypothesis* stipulates that a tax cut raises the budget deficit but leaves the external deficit unchanged, because rational, infinitely-lived consumers save the full amount of the tax cut in anticipation of future tax increases. The authors' OLG model nests both hypotheses as special cases: Ricardian equivalence holds when the birth rate $p = 0$ (representative agent) and the forecasting channel parameter $\pi_{12} = 0$, while the twin deficit hypothesis holds when $p > 0$ or $\pi_{12} \neq 0$.

2.2 The Blanchard (1985) OLG Model

The structural model is a small open economy populated by overlapping generations of consumers, following Blanchard (1985). The birth rate is p and survival probability is $(1 - p)$. At period t , the size of a cohort born at $s \leq t$ is $P_{s,t} = p(1 - p)^{t-s}$, with total population normalised to $P = 1$. When $0 < p < 1$, there are overlapping generations; when $p = 0$, the economy has a representative infinitely-lived consumer.

Each consumer from cohort s maximises expected discounted utility over a consumption stream:

$$\max E_t \left[\sum_{j=0}^{\infty} \left(\frac{1}{1 + \rho} \right)^j U(C_{s,t+j}) \right],$$

subject to the budget constraint:

$$A_{s,t+1} = (1 + \bar{\rho})(A_{s,t} + L_{s,t} - T_{s,t} - C_{s,t}),$$

where $A_{s,t} = B_{s,t} + F_{s,t}$ is non-human wealth (government plus foreign bonds), $T_{s,t}$ is lump-sum taxes, $L_{s,t}$ is stochastic (labour plus dividend) income, and the gross return on wealth is $(1 + \bar{\rho}) = (1 + r)/(1 - p)$ due to annuity arrangements with insurance firms.

Firms choose employment and investment to maximise the present value of dividends, with production $Y_t = Z_t K_t^\alpha N_t^{1-\alpha}$, capital accumulation $K_{t+1} = I_t + (1 - \delta)K_t$, and net output defined as $Q_t = Y_t - I_t - G_t$.

2.3 Budget Deficit, External Deficit, and Current Account

The government budget constraint is $B_{t+1} + B_{t+1}^* = (1 + r)(B_t + B_t^* + G_t - T_t)$, where B_t^* is government bonds held by foreigners and G_t is government spending. The *budget deficit* is:

$$D_t = \left(\frac{r}{1 + r} \right) (B_t + B_t^*) + G_t - T_t,$$

comprising debt service and the primary deficit. The *current account* is:

$$CA_t = \left(\frac{r}{1 + r} \right) (F_t - B_t^*) + Q_t - C_t,$$

where $F_t - B_t^*$ is net foreign assets and $Q_t - C_t$ is the trade balance. The *external deficit* is defined as $-CA_t$.

2.4 The Optimal Current Account Rule

Assuming quadratic preferences and solving the consumer's problem, then aggregating across cohorts and substituting into the current account equation, the authors derive the optimal rule:

$$CA_t = - \left(\frac{p}{1+r} \right) A_t - \left(\frac{\rho-r}{\rho} \right) D_t + x_t, \quad (1)$$

where x_t is the *adjusted current account*—a purely forward-looking component:

$$x_t = - \sum_{j=1}^{\infty} \left(\frac{1}{1+\rho} \right)^j E_t \left[\Delta Q_{t+j} + \left(\frac{\rho-r}{\rho} \right) \Delta D_{t+j} \right]. \quad (2)$$

Equation (1) shows that the current account depends on wealth A_t , the budget deficit D_t (scaled by the fraction of the tax burden shifted to unborn generations), and the forward-looking adjusted current account. Equation (2) states that x_t depends on expected future changes in net output ΔQ_{t+j} and future changes in the budget deficit ΔD_{t+j} . These two equations together embed both the *demographic channel* and the *forecasting channel*.

2.5 The Demographic Channel

The demographic channel operates via the OLG structure and is controlled by the birth rate p . When $0 < p < 1$, consumers can shift a fraction $(\rho-r)/\rho$ of their tax burden to unborn generations. A tax cut creates a positive wealth effect—consumers perceive an increase in lifetime resources—which raises consumption, lowers saving, and deteriorates the current account.

Formally, the budget deficit D_t enters the current account rule (1) directly via the term $-[(\rho-r)/\rho]D_t$. The net effect of an increase in the budget deficit is always to reduce the current account at impact, because the direct wealth effect dominates the indirect effect through adjusted current account dynamics. In simulations (Table 1 of the paper), the correlation between budget and external deficit ratios is positive regardless of the autoregressive parameter π_{22} , as long as $p > 0$.

When $p = 0$ (representative agent), $\rho = r$ and the budget deficit disappears entirely from the current account equation, yielding Ricardian equivalence as in [Key Point 2.1](#).

2.6 The Forecasting Channel

The forecasting channel is less standard and operates via the dynamic structure of the law of motion for forcing variables. It is controlled by the parameter π_{12} , which governs whether changes in the budget deficit Granger-cause changes in net output.

When $\pi_{12} \neq 0$, the budget deficit contains information useful for forecasting future net output. Even when $p = 0$ (no demographic channel), the budget deficit affects the current account *indirectly* through the adjusted current account solution $x_t = \theta_1 \Delta Q_t + \theta_2 \Delta D_t$, where $\theta_2 \neq 0$ as long as $\pi_{12} \neq 0$.

The direction of the forecasting channel is ambiguous and depends on the sign of π_{12} :

- If $\pi_{12} > 0$: a budget deficit increase signals future increases in net output (e.g., via military spending improving technology, or the “starve-the-beast” mechanism reducing future G_t). The resulting positive wealth effect raises consumption and worsens the current account $\rightarrow$ *positive comovement*.
- If $\pi_{12} < 0$: a budget deficit increase signals future *reductions* in net output (e.g., fiscal irresponsibility signalling higher future government spending). The negative wealth effect lowers consumption and improves the current account $\rightarrow$ *negative comovement*.

Simulations in Table 1 of the paper confirm this sign-dependence: correlations are positive for positive π_{12} and negative for negative π_{12} .

2.7 The VECM Representation and Empirical Strategy

To take the model to data, the authors generalise the law of motion for forcing variables into a vector error correction model (VECM). The baseline law of motion $H_{t+1} = \Pi H_t + V_{t+1}$ is improved in two ways. First, an *error correction term* βy_t is added, where $y_t = (Q_t \ D_t)'$, to capture the long-run cointegrating relationship between the budget deficit and net output (consistent with government solvency and Bohn's 1998 finding of debt-to-output mean reversion). Second, the *adjusted current account* x_t is included as an additional variable to summarise agents' forward-looking information set.

The resulting unrestricted VECM is:

$$\mathbf{X}_{t+1} = \Gamma \mathbf{X}_t + \beta \mathbf{y}_t + \mathbf{U}_{x,t+1},$$

where $\mathbf{X}_t = (\Delta Q_t \ \Delta D_t \ x_t)'$. The restricted VECM $\hat{\mathbf{X}}_{t+1} = \Gamma_x \hat{\mathbf{X}}_t + \beta_x \mathbf{y}_t + \hat{\mathbf{U}}_{x,t+1}$ embeds all the structural model's restrictions through the reduced-form parameter vector ν , which is a complex function of the estimated law-of-motion parameters, ρ , and r .

An analogous VECM is derived for aggregate wealth dynamics. All empirical results are reported for both the unrestricted and restricted VECMs.

2.8 Birth Rate Estimation and Model Validation

The restricted reduced forms involve two parameters: the interest rate $r = 0.04$ (calibrated) and the birth rate p (estimated). The birth rate is estimated using orthogonality restrictions implied by the model: under the null that the model holds, the variable $\varepsilon_t = x_t - (1 + \rho)x_{t-1} - \Delta Q_t - [(\rho - r)/\rho]\Delta D_t$ must be orthogonal to all lagged variables. The authors perform a grid search and also obtain GMM estimates $\hat{p}$.

For all 16 countries, the actual annual birth rates $\wp$ (ranging from 1.0% for Germany to 1.5% for Australia, Canada, and the U.S.) fall inside the 1% significance-level bounds. The overidentifying restrictions related to the GMM estimates are never rejected. Furthermore, comparing the actual adjusted current account x_t to its fitted counterpart $\hat{x}_t$ from the restricted reduced form confirms an overall good fit—particularly strong for Finland, Germany, Sweden, and Switzerland.

2.9 Identification of Tax Shocks

The identification of orthogonal tax shocks is achieved via a Cholesky decomposition applied to the VECM innovations $\mathbf{U}_{x,t}$. The ordering is: (1) net output, (2) budget deficit, (3) adjusted current account. This decomposition is consistent with the structural model:

- Equation (21): $\lambda_{11}^* u_{Q,t} = \xi_{Q,t}$. Net output innovations depend only on the output shock $\xi_{Q,t}$ —consistent with the model's property that net output is unaffected contemporaneously by tax policies.
- Equation (22): $\lambda_{22}^* u_{D,t} = -\lambda_{21}^* u_{Q,t} + \xi_{T,t}$. Budget deficit innovations depend on the output innovation (capturing automatic stabilisers) and an orthogonal tax shock $\xi_{T,t}$. This purges the endogenous business-cycle component from net tax revenues.
- Equation (23): $\lambda_{33}^* u_{x,t} = -\lambda_{31}^* u_{Q,t} - \lambda_{32}^* u_{D,t} + \xi_{x,t}$. The adjusted current account innovation depends on both prior innovations and an additional shock.

Robustness is verified using the alternative Blanchard and Perotti (2002) identification scheme.

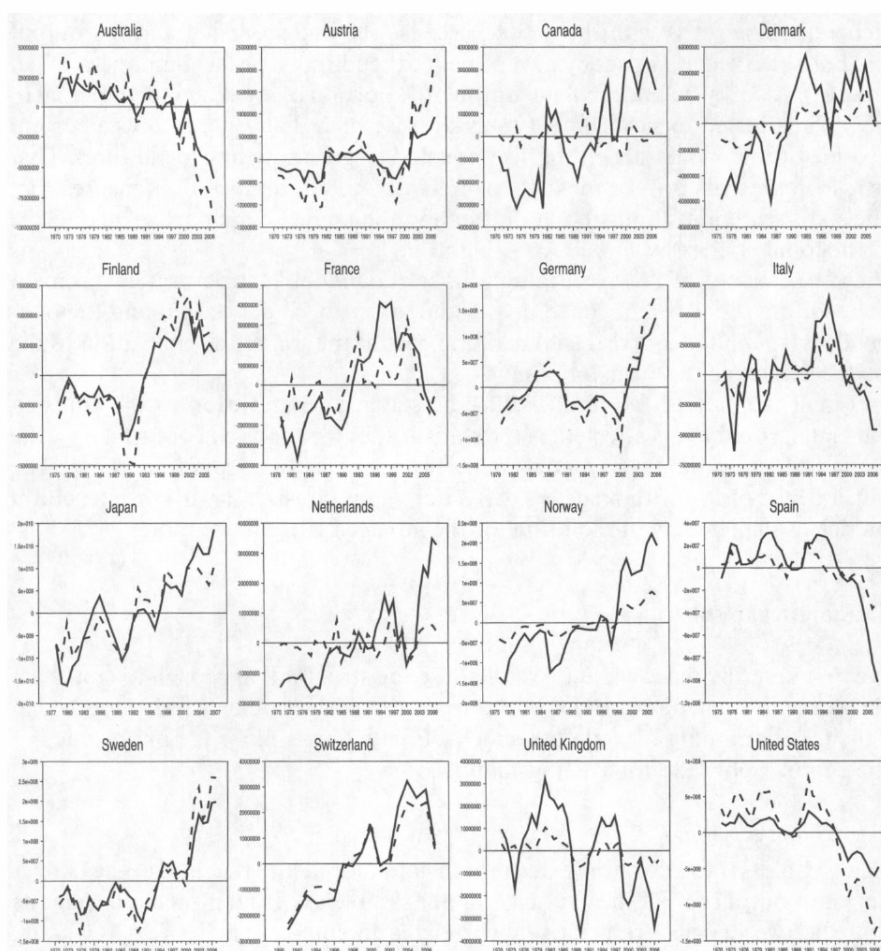


FIGURE 2 Actual and predicted adjusted current account

NOTES: The solid lines show the actual adjusted current account x_t . The dashed lines display the fitted values of the adjusted current account $\hat{x}_t$ generated by the restricted VECMs. All computations used the actual values of the birth rate and the unrestricted law of motion (FULL).

The above decomposition is consistent with the structural model of section 2. For example, equation (21) implies that the statistical innovation to net output $u_{Q,t}$ depends only on the orthogonal shock $\xi_{Q,t}$. This is consistent with the model, because net output is unaffected at impact by tax policies and current account movements (see the technical appendix, A.2).

Equation (22) implies that the statistical innovation to the budget deficit $u_{D,t}$ is affected by the statistical innovation to net output $u_{Q,t}$ and an orthogonal shock $\xi_{T,t}$. The dependence of $u_{D,t}$ with respect to $u_{Q,t}$ reflects changes in

Figure 1: Actual (solid) and predicted (dashed) adjusted current account. The model fits the data well across most countries, confirming the validity of the structural restrictions. (Reproduced from Figure 2 of the paper.)

2.10 Unconditional Correlations: No Clear Pattern

Before applying the structural approach, the authors document that the raw data show no consistent pattern. Unconditional correlations between budget deficit and external deficit ratios range from -0.818 (Spain) to $+0.648$ (Norway), with eight countries showing positive and eight showing negative correlations. Many are statistically insignificant. This heterogeneity motivates the structural exercise: isolating tax shocks is essential because multiple shocks simultaneously drive the two deficits in potentially opposing directions.

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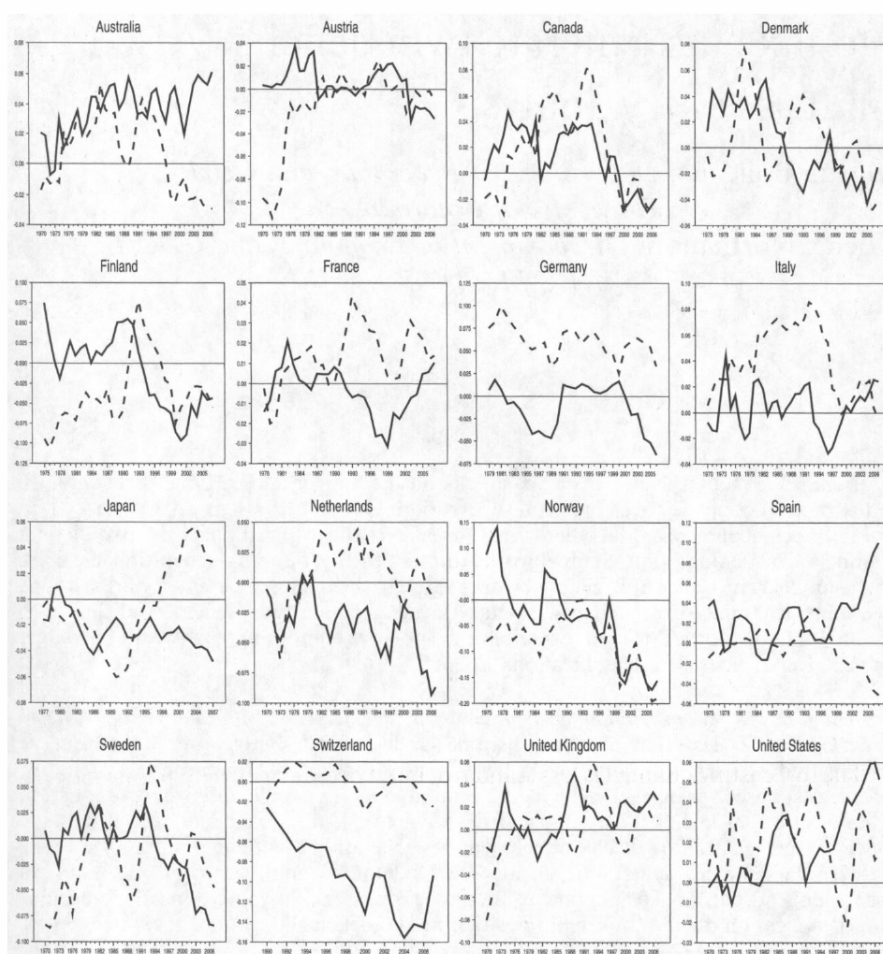


FIGURE 1 Budget and external deficit ratios

NOTE: The solid lines show the external deficit to output ratios, while the dashed lines show the budget deficit to output ratios.

Muller 2010; Chinn 2005; Corsetti and Muller 2006 and 2008; Kim and Roubini 2008; Monacelli and Perotti 2010). One view is that large budget deficits reduce aggregate saving and generate external deficits. The empirical support for this view in international data, however, is slim. As shown in figure 1, the budget and external deficits series do not display a homogeneous pattern of positive comovements among industrialized countries. In part, the lack of homogeneous evidence occurs because the raw data jointly capture the effects of several shocks. These shocks may simultaneously push the budget and external deficits in different

Figure 2: Budget deficit (dashed) and external deficit (solid) ratios to output for 16 countries. The raw data display diverse patterns with no uniform positive comovement. (Reproduced from Figure 1 of the paper.)

2.11 Conditional Correlations: Strong Twin Deficits

Conditioning on only tax shocks, the correlations change dramatically. Using the *unrestricted VECM* with Cholesky identification, the conditional correlations are positive and statistically significant for 15 of 16 countries (all but Germany at 10%), ranging from 0.401 to 0.904. Dropping the error correction term (VAR specification) or using the Blanchard–Perotti identification yields similarly positive results.

Using the *restricted VECM* (full structural model), the conditional correlations are even

larger—ranging from 0.386 (Austria) to 0.995 (Sweden)—and statistically significant at any conventional level for all 16 countries. This uniform pattern of large, positive correlations provides strong evidence for the twin deficit hypothesis when the analysis is conditioned on tax shocks.

2.12 Dynamic Responses to a Tax Cut

Figure 3 of the paper shows the impulse responses of both deficit levels to a unit tax-cut shock, computed from the restricted reduced forms. By construction, the budget deficit rises by one unit on impact for all countries and remains persistently positive (except for gradual declines in the Netherlands and Sweden).

The external deficit displays a large positive impact response for most countries (notably above unity for Austria and France). Exceptions include a negative impact response for Japan and Spain, and a near-zero response for Italy and Switzerland. However, for these countries, the external deficit rises sharply within a few years of the shock. The positive responses of both deficits are highly persistent, which explains the large positive conditional correlations documented in [Key Point 2.11](#).

2.13 Variance Decomposition and Debt Accumulation

The tax shock accounts for a very large fraction of the forecast error variance of the budget deficit at the one-year horizon (68–100% depending on the country), declining somewhat at longer horizons due to the model dynamics. More importantly, the tax shock also explains a sizeable fraction of the external deficit’s forecast error variance for most countries. For example, at the 10-year horizon, it explains 88% in France, 48% in Canada and Denmark, and 86% in Spain, though it explains relatively little in Austria (3%), Germany (3%), and Switzerland (7%).

Cumulating the deficit responses yields the public and external debt paths. After a decade, a unit tax cut generates substantial public debt increases (5.6 to 13.8 currency units) for all countries, and positive external debt accumulation for all countries (1.1 to 7.3 currency units). In Canada, a \$1 tax cut produces approximately \$11 of public debt and \$3 of external debt after 10 years.

2.14 Disentangling the Demographic and Forecasting Channels

To isolate each channel, the authors re-estimate the structural model under two parameterisations:

- **Forecasting only (FCST):** The demographic channel is shut down by setting $p = 0.001$ (nearly zero). This eliminates the intergenerational wealth effect.
- **Demographic only (DEMO):** The forecasting channel is shut down by assuming univariate AR processes for ΔQ_t and ΔD_t (making the VAR matrices lower triangular and removing the error correction term). This ensures the budget deficit contains no information useful for forecasting net output.

The results reveal complementary roles. Under FCST, conditional correlations are positive and significant for 12 of 16 countries (insignificant for Finland, Germany, and Switzerland). The dynamic responses of the external deficit have similar *shapes* to the full model but much smaller *amplitudes*. Under DEMO, correlations are uniformly positive, large (0.885 to 0.991), and significant for all 16 countries. The responses are always positive and persistent, but lack the nuanced shapes produced by the forecasting channel.

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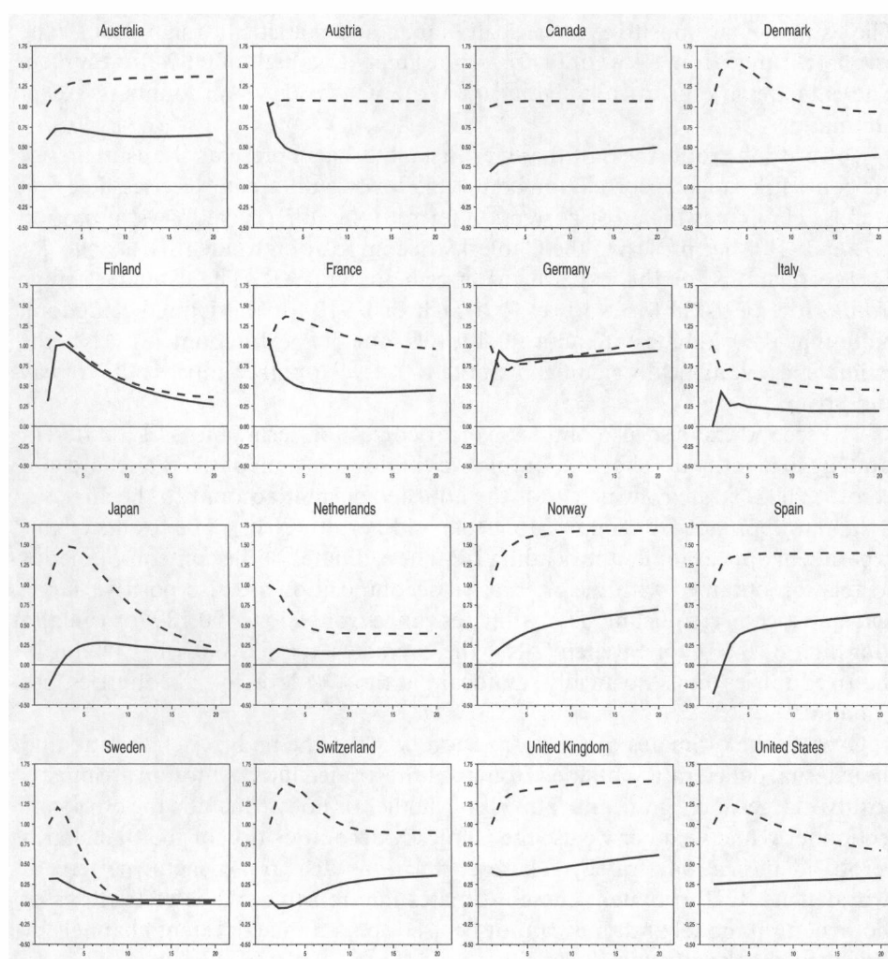


FIGURE 3 Dynamic responses of deficit levels: FULL

NOTES: The solid lines correspond to the dynamic responses of the external deficit level and the dashed lines correspond to the dynamic responses of the budget deficit level. All computations are based on the actual values of the birth rates and the unrestricted law of motion (FULL).

Thus far, the country estimates of the correlation between the budget and external deficits constructed from the restricted reduced forms support a general pattern of twin deficits. Before we study the importance of the different channels by which tax shocks affect the external deficit, it is worthwhile to understand the general mechanisms at play. Intuitively, the positive correlation between the budget and external deficits can be attributed to the positive responses of both the budget and external deficits to a tax cut over most horizons. Accordingly,

Figure 3: Dynamic impulse responses of external deficit (solid) and budget deficit (dashed) levels to a unit tax cut, from the full structural model (FULL). (Reproduced from Figure 3 of the paper.)

2.15 Model Limitations and Extensions

The authors acknowledge several limitations. First, the model assumes that net output is unrelated to consumers' saving decisions, which holds only with exogenous interest rates and no wealth effects on labour supply. Introducing a debt-elastic interest rate or endogenous labour supply with standard preferences would break this dichotomy. Second, the model uses lump-sum taxes, whereas real-world taxes are distortionary. Third, modelling all 16 countries (including the U.S.) as small open economies may be inappropriate for the larger economies. The authors

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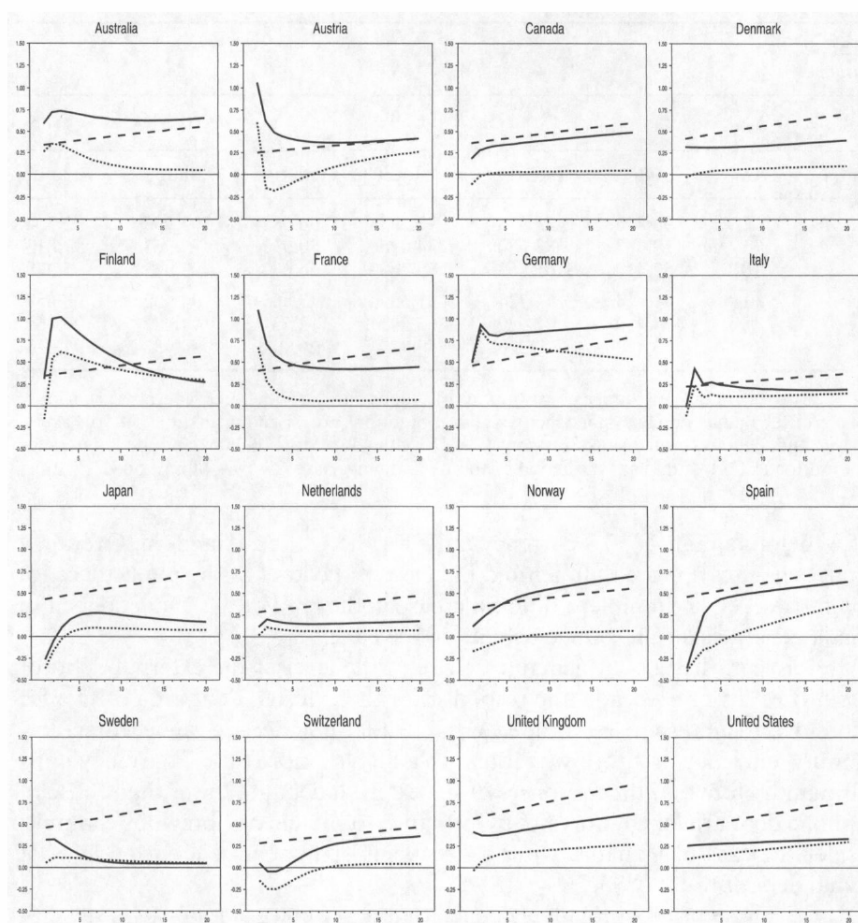


FIGURE 4 Dynamic responses of deficit levels: DEMO vs. FCST

NOTES: The solid lines correspond to the dynamic responses of external deficit level for the version with the actual birth rate and the unrestricted law of motion (FULL). The dotted lines show the dynamic responses of external deficit level for the version with a birth rate of 0.001 and the unrestricted law of motion (FCST). The dashed lines represent the dynamic responses of external deficit level for the version with the actual birth rate and univariate AR laws of motion (DEMO).

small differences in the responses of the budget deficit to the tax shock, but the responses remain positive at all horizons.

For tables 3, 4, and 5, as well as figure 4, the results under forecasting (FCST) are computed for a version of the full structural model that closes the demographic channel. For this, the birth rate is set to $p = 0.001$ for all countries. Although this value is not strictly null, it is small enough to ensure that the demographic channel is negligible. With this parameterization, consumers find

Figure 4: External deficit impulse responses under FULL (solid), FCST (dotted), and DEMO (dashed). The forecasting channel determines the shape; the demographic channel determines the amplitude. (Reproduced from Figure 4 of the paper.)

suggest that relaxing these assumptions—particularly allowing for endogenous interest rates, distortionary taxes, and heterogeneous country sizes—would be valuable extensions.

3 Logical Chains

3.1 From Tax Cut to Twin Deficits via the Demographic Channel

Step 1. Tax cut reduces government revenue: The budget deficit D_t increases because $G_t - T_t$ rises.

Explanation: With government expenditures G_t held constant, a reduction in lump-sum taxes T_t mechanically increases the primary deficit and hence D_t .

Step 2. Overlapping generations allow intergenerational tax shifting: When $p > 0$, current consumers can shift a fraction $(\rho - r)/\rho$ of the future tax burden to unborn generations.

Explanation: In the Blanchard (1985) OLG framework (see [Key Point 2.2](#)), current consumers discount future taxes at the rate $\rho > r$, so a tax cut is not fully offset by expected future taxes within their lifetimes.

Step 3. Positive wealth effect on current consumers: Current consumers perceive higher lifetime wealth A_t .

Explanation: The budget deficit enters the current account rule (1) via $-(\rho - r)/\rho D_t$, reflecting that consumers keep a share of the tax cut as a net wealth gain.

Step 4. Consumption rises, saving falls: Higher perceived wealth increases aggregate consumption C_t and reduces aggregate saving.

Explanation: Under permanent-income logic with quadratic preferences, consumption is proportional to perceived wealth.

Step 5. External deficit increases: Lower national saving, for unchanged investment and government spending, worsens the current account: CA_t falls and the external deficit rises.

Explanation: By the national accounting identity, a fall in saving relative to investment must be financed by foreign borrowing.

Conclusion. A tax cut simultaneously raises both the budget and external deficits—the twin deficit outcome—via the wealth effect inherent in the OLG structure.

Explanation: Steps 1–5 form a complete causal chain from tax cut $\rightarrow$ budget deficit $\rightarrow$ wealth effect $\rightarrow$ consumption boom $\rightarrow$ external deficit, as summarised in [Key Point 2.5](#).

3.2 From Tax Cut to Twin Deficits via the Forecasting Channel

Step 1. Tax cut increases the budget deficit: As in [Logical Chain 3.1](#), the budget deficit D_t rises.

Explanation: Same mechanism as Step 1 of the demographic chain.

Step 2. Budget deficit contains information about future net output: When $\pi_{12} \neq 0$ in the law of motion, changes in D_t Granger-cause changes in net output Q_t .

Explanation: Several mechanisms justify this link. For instance, budget elements (military spending, political parties) may predict future productivity (Evans 1992; Hall 1988). The “starve-the-beast” mechanism suggests that current tax cuts signal future spending cuts. See [Key Point 2.6](#).

Step 3. Rational agents update forecasts of future net output: If $\pi_{12} > 0$, the rise in ΔD_t signals higher future ΔQ_{t+j} .

Explanation: Agents use all available information—including the current budget deficit—to form rational expectations of future forcing variables in the adjusted current account formula (2).

Step 4. Expected higher future output raises perceived wealth: The present value of expected future income increases.

Explanation: The adjusted current account $x_t = \theta_1 \Delta Q_t + \theta_2 \Delta D_t$ captures this effect; with $\pi_{12} > 0$, we have $\theta_2 < 0$, meaning a rise in ΔD_t reduces x_t (worsens the current account).

Step 5. Consumption rises, current account deteriorates: Same wealth-consumption logic as the demographic channel.

Explanation: Higher wealth induces higher consumption and lower saving, worsening the external balance.

Conclusion. **When the budget deficit signals future output increases ($\pi_{12} > 0$), the forecasting channel reinforces twin deficits; when it signals future output decreases ($\pi_{12} < 0$), it works against them.**

Explanation: The direction of the forecasting channel is empirically determined. The paper finds that, in practice, the estimated π_{12} values produce positive correlations for most countries.

3.3 From Raw Heterogeneity to Structural Homogeneity

Step 1. Raw data show heterogeneous deficit correlations: Unconditional correlations range from -0.818 (Spain) to $+0.648$ (Norway).

Explanation: Multiple shocks (tax, spending, productivity) simultaneously affect both deficits in potentially offsetting ways. See [Key Point 2.10](#).

Step 2. Structural model permits isolation of tax shocks: The VECM framework with Cholesky identification strips out the endogenous business-cycle component of the budget deficit, leaving only exogenous tax innovations $\xi_{T,t}$.

Explanation: As described in [Key Point 2.9](#), the Cholesky ordering ensures that the tax shock is orthogonal to contemporaneous output shocks.

Step 3. Conditional correlations are uniformly positive and large: Across all 16 countries, the conditional correlations range from 0.386 to 0.995 .

Explanation: Once the confounding effects of other shocks are removed, the underlying twin-deficit relationship emerges clearly. See [Key Point 2.11](#).

Conclusion. **The apparent heterogeneity in raw data is a consequence of omitted-variable bias; once only tax shocks are considered, the twin deficit hypothesis holds universally across industrialised countries.**

Explanation: Steps 1–3 demonstrate that the lack of evidence for twin deficits in raw correlations is a statistical artefact, not a refutation of the underlying economic relationship.

4 Key Terms & Definitions

Adjusted current account (x_t)

The purely forward-looking component of the optimal current account rule, defined as $x_t = -\sum_{j=1}^{\infty} (1 + \rho)^{-j} E_t[\Delta Q_{t+j} + ((\rho - r)/\rho) \Delta D_{t+j}]$. It captures agents' expectations of future changes in net output and the budget deficit. Under the model's solution, it reduces to $x_t = \theta_1 \Delta Q_t + \theta_2 \Delta D_t$.

Birth rate (p)

The fraction of the population born each period in the OLG model. When $p > 0$, overlapping generations exist and consumers can shift part of their tax burden to unborn generations. When $p = 0$, the model collapses to a representative agent with Ricardian equivalence. Actual annual birth rates in the sample range from 1.0% (Germany) to 1.5% (Australia, Canada, U.S.).

Budget deficit (D_t)

Defined as $D_t = [r/(1 + r)](B_t + B_t^*) + G_t - T_t$, comprising debt service payments plus the primary deficit. Measured empirically as the negative of the nominal budget surplus.

Cholesky decomposition

The identification strategy used to extract orthogonal tax shocks from the VECM residuals. The ordering (net output first, budget deficit second, adjusted current account third) ensures that the output innovation is not contaminated by tax policy, and that the tax shock $\xi_{T,t}$ is the portion of the budget deficit innovation orthogonal to output innovations—i.e., it purges the automatic-stabiliser component.

Current account (CA_t)

Measured as $CA_t = [r/(1 + r)](F_t - B_t^*) + Q_t - C_t$, where the first term is net income from foreign assets and the second is the trade balance. The external deficit is defined as $-CA_t$.

Demographic channel

The channel by which the budget deficit affects the external deficit through the OLG structure. Controlled by $p > 0$, it ensures that consumers can shift a fraction $(\rho - r)/\rho$ of the tax burden to unborn generations, creating a wealth effect. It always promotes *positive* comovements between the two deficits.

Error correction term

The term $\beta \mathbf{y}_t$ in the VECM law of motion, where $\mathbf{y}_t = (Q_t \ D_t)'$. It captures the long-run cointegrating relationship between the budget deficit and net output, consistent with government solvency. Its inclusion strengthens the forecasting channel by providing an additional mechanism through which the current budget deficit predicts future net output.

External deficit

The negative of the current account, $-CA_t$. It measures the extent to which the domestic economy borrows from abroad in period t .

Forecasting channel

The channel by which the budget deficit affects the external deficit through the informational content of the budget deficit for predicting future net output. Controlled by π_{12} in the VAR law of motion for forcing variables. It can produce either positive or negative comovements, depending on the sign of π_{12} .

Net output (Q_t)

Defined as $Q_t = Y_t - I_t - G_t$: output net of investment and government expenditures. In the model with inelastic labour and exogenous interest rate, net output depends only on exogenous variables (productivity Z_t , government spending G_t , and r) and is therefore unaffected by contemporaneous tax policy.

Ricardian equivalence

The hypothesis that a tax cut raises the budget deficit but leaves the external deficit unchanged, because forward-looking consumers fully save the tax cut to pay for expected future tax increases. In the paper's framework, it holds when $p = 0$ (representative agent) and $\pi_{12} = 0$ (no forecasting channel).

Twin deficit hypothesis

The hypothesis that a tax cut simultaneously raises both the budget and external deficits. In the paper's framework, it holds when $p > 0$ (demographic channel active) or $\pi_{12} \neq 0$ (forecasting channel active), such that the budget deficit transmits to the external deficit.

VECM (Vector Error Correction Model)

The econometric representation of the model's dynamics: $\mathbf{X}_{t+1} = \Gamma \mathbf{X}_t + \beta \mathbf{y}_t + \mathbf{U}_{x,t+1}$, where $\mathbf{X}_t = (\Delta Q_t \ \Delta D_t \ x_t)'$. The unrestricted version is estimated directly; the restricted version embeds all the structural model's cross-equation restrictions.